

Water Before Steam: Canals, Coal and the Making of Britain's Fossil Economy, 1700–1870

ABSTRACT

Why did the fossil economy begin in Britain rather than in the Netherlands, with Europe's densest waterways, or China, with coal and rivers in abundance? Terje Tvedt's answer, in this journal, was that Britain alone could engineer its water to deliver coal cheaply: water was the precondition for a coal economy. That is a claim about sequence, and Britain is where the annual series to test it exist together. Using British sectoral output for 1700–1870, a new canal-mileage series, installed horsepower and the Google Books corpus, we find two growth regimes: an aggregate acceleration from 1775–1792, as the canals were built, absorbed by population, and a per-capita acceleration from 1818 that belongs to steam. Canal mileage predicts coal output, not income. At the Maddison benchmarks the Dutch, Belgian, Chinese and Indian records are consistent with the rule that neither engineered water nor coal was sufficient alone.

Keywords: Industrial Revolution, canals, coal, water infrastructure, Great Divergence, Netherlands, China, energy transition, growth regimes

1. Introduction

“Why England and not China and India?” Terje Tvedt asked in this journal in 2010, and answered with water¹. Pomeranz had asked the same question of the Yangzi delta a decade earlier and answered with coal and colonies². The two answers are usually read as rivals. They need not be. In 1700 the parts of what became Britain's fossil economy were scattered across several economies, and none held them all. The Netherlands had the densest network of engineered waterways in Europe and no coal³. China had coal in quantity and the longest artificial waterway in the world, and the two lay a thousand kilometres apart⁴. India had coal in Bengal and its cotton industry on the western and south-eastern coasts, joined by rivers whose flow followed the

monsoon⁵. Belgium had coal under Hainaut and Liège and would acquire the canals to move it in 1818 and 1832, after which its income per head grew faster than Britain's⁶. Britain had coal, rivers with the gradient to drive mills and the flow to float barges, and a legal form, the canal Act, that let private capital join the two. By 1830 no economy had gone further in running on mineral fuel moved by water.

Tvedt's claim is that the water system was the *precondition* for the coal economy, not a rival to it. This paper takes that claim literally and tests it. A precondition is a claim about order and dependence, and it makes predictions that a rival-cause claim does not. It predicts that the sectors served by canals accelerated before steam was a significant source of power; that coal output responded to the canal network; that steam capacity followed coal; and that the sectors canals did not serve, and the outcome canals could not directly raise, did not move. Each prediction is about sequence within one economy, and sequence can be tested only where the relevant annual series exist together. The Netherlands and Sweden have annual output from before 1700, but only Britain also has canal mileage and installed power year by year over a period in which all three changed. The comparative cases can then be read through its result.

The test has to reconcile two clocks that the literature has kept apart. The steam clock starts in the 1780s with Watt's rotative engine and reads industrialisation as the arrival of a new prime mover and of growth in income per head, which barely moved before 1830⁷. The canal clock starts in 1761, when the Duke of Bridgewater's canal halved the price of coal in Manchester, and is calibrated to tonnage, prices and towns⁸. Setting the two side by side on annual data produces three findings.

First, Britain had two growth regimes, not one. Using the annual output series of Broadberry, Campbell, Klein, Overton and van Leeuwen, we find trend breaks between 1775 and 1792 in total output, industry, services, coal, iron and population, and a break in 1818 in income per head. In the first regime the growth of aggregate output nearly tripled and the growth of population quadrupled, while income per head grew no faster than it had since 1700 and, between 1760 and 1790, slower. Population absorbed the acceleration. Both clocks are right, about different variables.

Second, the canal network is a dose that predicts the right things. We assemble a year-by-year series of canal miles from the completion dates of 155 English and Welsh canals and corroborate its timing with 152 parliamentary authorisations recovered from Priestley's 1831 survey.

Cumulative canal mileage predicts coal output over the following five to twenty years and it predicts population; it predicts neither income per head nor agricultural output, and the coal result survives a quadratic trend, first differences and two predetermined versions of the dose. Steam horsepower, in turn, raises income per head only after 1830. In the Google Books British corpus the same order appears, to within a decade: “coal barge” and “coal wharf” reach a quarter of their mid-century frequency around 1781 and 1800; “steam engine” and “steam power” around 1808 and 1826.

Third, the comparative cases behave as a precondition thesis says they should, and the standard cross-country test cannot see it. Between the Maddison benchmark years 1700 and 1820, before steam was a third of British power, British output tripled while Dutch output grew by 9 per cent: water transport without coal, and without falling water to drive mills, did not compound. Belgium, whose coalfield canals opened in 1818 and 1832, grew per head by 8 per cent between 1700 and 1820 and by 82 per cent in the next fifty years, more than Britain. China nearly trebled its population and lost income per head. A cross-country difference-in-differences on income per head, meanwhile, locates a British take-off in 1807, which is the year the Dutch economy lost 44 per cent of its income under French occupation while Britain held level.

The contribution is to four literatures. To global history and the Great Divergence debate, we give Tvedt’s thesis a quantitative test and restate it as a rule the comparative cases can be checked against: neither engineered water nor coal was sufficient, and the economies that had one without the other did not industrialise until they had both. To the transport-revolution literature, we add an annual national dose series and a sequencing test that connects canals to the energy transition rather than to urban growth, the channel recently quantified by Alvarez-Palau and co-authors⁹. To the energy-history debate among Wrigley, Allen, Pomeranz and Malm, we add the observation that the escape from the Malthusian ceiling was under way, in aggregate, before steam, in Wrigley’s advanced organic economy. To the practice of historical difference-in-differences¹⁰, we add a worked example of how a control group’s catastrophe becomes a treatment group’s miracle.

The next section places the argument in the literature; the following two describe the data and report the results, British first and comparative second; the discussion returns to Tvedt’s question, and the limitations, which are real, come before the conclusion.

2. Water, coal and the divergence debate

2.1 The transport revolution and its canals

The transport revolution of eighteenth-century Britain is well documented and, until recently, poorly measured. Between the Bridgewater Canal of 1761 and the opening of the Liverpool and Manchester Railway in 1830, Parliament authorised roughly 165 canal companies and the navigable network grew from about 1,400 miles of improved river to some 4,000 miles of river and canal¹¹. The coal trade itself was older and larger than the canals: the coastal trade from the Tyne and Wear to London was already the largest bulk-freight flow in Europe in 1700, and Britain's coal output had been rising for two centuries before the first canal was cut¹². What canals changed was inland access. Priestley's survey of 1831 opens its account of the Bridgewater Canal by noting that the Duke's "primary object" was "to open his valuable collieries at Worsley, and to supply the town of Manchester with coal, at a much cheaper rate"¹³. Turnbull's regional study made the same point: canals were built where coal was, carried coal above all else, and the counties that acquired them saw coal and population grow faster¹⁴.

Quantitative work has caught up in the last decade. Bogart's survey collects the mileage, cost and freight-rate evidence and concludes that inland freight costs fell by half or more between 1700 and 1830¹⁵. Alvarez-Palau, Bogart, Satchell and Shaw-Taylor estimate that transport costs relative to producer prices fell by nearly 75 per cent between 1680 and 1830 and that, without that fall, inland towns would have been 20 to 25 per cent smaller in 1841¹⁶. Their outcome is urban population. Ours is the energy system. The two are complementary: towns grew where coal could be delivered, and coal could be delivered where water had been engineered.

2.2 Energy, the organic economy and the Great Divergence

The energy literature starts from a different place. Wrigley's organic economy is bounded by the annual product of the land; sustained growth required a "mineral-based energy economy" in which coal replaced wood as a source of heat and, much later, water and wind as a source of power¹⁷. Wrigley is careful that the first transition was under way from the sixteenth century and that the eighteenth-century economy was an advanced organic one, burning mineral heat while its power remained organic; Warde's energy accounts confirm the chronology¹⁸. The steam clock belongs to Wrigley's readers more than to Wrigley. Allen's induced-innovation account makes

cheap British coal and dear British labour the reason the steam engine was worth inventing in Britain and nowhere else¹⁹. Pomeranz gives coal a starring role in the Great Divergence, alongside the ghost acres of the Americas²⁰. Clark and Jacks and Fernihough and O'Rourke have found that proximity to coal mattered a great deal for where Europe industrialised²¹. Malm's *Fossil Capital* accepts coal's centrality but denies that its adoption was driven by thermodynamic superiority or scarcity of water power: steam won because it freed capital from the riverbank and allowed mills to be sited where labour could be disciplined²².

What these accounts share is a clock calibrated to steam and coal combustion. Tvedt's intervention is to move the clock upstream. In "Why England and not China and India?", he argues that Britain's distinctive endowment was not coal as such, which China had in abundance, but a hydrological regime of gentle gradients, steady flow and navigable rivers that could be engineered into a national network of canals and water-powered mills at low cost²³. The monsoonal rivers of Asia could not. On this reading the water system is the precondition: it created the market integration, the cheap coal at the point of use, and the water-powered factory template on which steam later built. The Asian side of the comparison has its own literature. Elvin's high-level equilibrium trap describes a Chinese economy that had exhausted its organic frontier without reaching its mineral one, and Pomeranz locates the reason in geography: China's coal lay in the north-west, far from the Jiangnan core²⁴. Parthasarathi makes a parallel case for India, whose coal in Bengal lay far from the cotton regions and whose rivers ran with the monsoon, and argues that endowments and pressures, not institutions or culture, pushed Britain to coal²⁵. Tvedt's evidence is comparative and qualitative. This paper asks whether the annual British record supports the sequence he proposes, and then reads the comparative cases against what it finds.

A precondition is not a rival cause. Coal remains the fuel of the second regime and steam its engine. The claim is that neither would have arrived when and where it did without the first regime, and that the first regime is visible in the data if one looks at the variables canals could move: coal, aggregate output, and the number of people the economy could feed.

2.3 Two chronologies of growth

The macroeconomic chronology has moved decisively towards gradualism. Crafts, and Crafts and Harley, cut the growth rates of the 1780–1830 period that Deane and Cole had estimated and showed that income per head grew slowly until well into the nineteenth century²⁶. Broadberry, Campbell, Klein, Overton and van Leeuwen’s reconstruction of British output from the medieval period confirms the picture: annual growth of GDP per head averaged around 0.3 per cent from 1700 to the 1820s and only then accelerated²⁷. Crafts’s growth-accounting of steam finds its contribution to labour-productivity growth trivial before 1830 and dominant only after 1850²⁸. On Kanefsky’s estimates of installed power, steam did not overtake water until the 1830s²⁹.

Gradualism in income per head has been read as a verdict against any eighteenth-century take-off. It is not. The same Broadberry series show total output and population doubling between 1760 and 1830. A growth regime in which aggregate output accelerates while income per head does not is exactly what Malthus described and exactly what an infrastructure shock in an organic economy should produce: more mouths fed, not richer mouths. The distinction between the aggregate and the per-capita clock is, we argue, the key to reconciling the canal historians with the cliometricians.

2.4 Text as data and historical difference-in-differences

Two methodological literatures frame our tools. Michel et al. introduced the Google Books corpus as a quantitative record of culture, and Pechenick, Danforth and Dodds documented its limits: it over-weights technical publishing and records what was printed rather than said³⁰. For infrastructure vocabulary that bias is a feature. The historical difference-in-differences literature has grown cautious about serial correlation, heterogeneous effects and pre-trends³¹; we add a more elementary hazard, a control group hit by a war the treatment group escaped, whose economic consequences Crouzet described sixty years ago³².

3. Data and empirical strategy

3.1 British output, population and power

Annual real output for Great Britain by sector, 1700–1870, comes from Broadberry, Campbell, Klein, Overton and van Leeuwen as published in the Bank of England’s *A Millennium of Macroeconomic Data for the UK*, version 3.1³³. We use total GDP, agriculture, industry and services, and from the industrial-production tables the physical output indices for coal, iron and textiles. Population of Great Britain is from the same source, based on Wrigley and Schofield’s reconstitution to 1801 and the censuses thereafter. GDP per head is total output divided by population. The series are indices with 1700 = 100.

Installed stationary power is from Kanefsky’s estimates as tabulated by Crafts, at the benchmark years 1760, 1800, 1830 and 1870, reproduced in Table 4³⁴. We interpolate log-linearly between the four benchmarks, wind included. The interpolated series are smooth by construction, and we treat inference that depends on their year-to-year variation with corresponding caution.

3.2 The canal network as a dose

A single treatment date cannot represent the canal network. It was built over sixty years in two waves, and a binary indicator cannot distinguish the Grand Cross of the 1770s from the mania canals completed between 1790 and 1816. We therefore construct a continuous dose, which we call the canal stock: cumulative canal miles in operation by year.

The primary series is assembled from the completion year and length of 155 canals in England and Wales, taken from the reference table of British canals maintained on Wikipedia and retrieved on 3 September 2026; we include every entry with a stated length and opening year, and date staged openings to full completion, which biases the series late by a few years. The 155 canals total 2,967 miles, of which 2,320 were open by 1830. The series excludes river navigations improved before 1700, which were the bulk of the network’s initial 1,400 miles, so it measures additions to the network rather than its level; for the growth regressions this is the relevant quantity. The Cambridge Group’s dynamic GIS puts the England and Wales network at about 1,400 miles in 1760 and 4,000 in 1835, most of the growth canal³⁵; our 2,102 miles over

1760–1830 are four-fifths of that increase, the rest post-1830 completions, branches and river work we omit. Hadfield’s roughly 165 canal companies authorised between 1761 and 1830 counts undertakings, some of which built no canal, and is not comparable³⁶.

We corroborate the timing with an independent source. Priestley’s *Historical Account of the Navigable Rivers, Canals, and Railways throughout Great Britain* records the Acts that authorised each navigation with regnal year and date of Royal Assent³⁷. From the digitised 1831 text we identify 325 entries by their capitalised headings and recover a first-authorisation year for 152, including 70 canals authorised between 1760 and 1829, thirty of them in the 1790s; the rest are lost to the quality of the optical character recognition, and lengths could not be recovered at all. The authorisation series leads the completion series by roughly a decade, as it should, and both show the same two waves.

3.3 Comparative benchmarks

Cross-country GDP per head and population are from the Maddison Project Database 2023³⁸. The series labelled GBR is the United Kingdom including Ireland; we note where this matters. For Britain, France, the Netherlands, Sweden, Germany, Spain and Portugal the series is annual from 1700; for Belgium, China and India it is interpolated between benchmarks. Population in the Maddison database is a benchmark series before 1820, so the comparative cases are read at the benchmark years 1700, 1820 and 1870, alongside what the national literatures record about fuel, gradient and waterways: de Vries and de Zeeuw for the Netherlands, Mokyr for Belgium, Pomeranz and Elvin for China and Parthasarathi for India³⁹.

3.4 Text data

Word and phrase frequencies are from the Google Books Ngram corpus, British English 2019 edition, 1700–1900, retrieved with the viewer’s three-year smoothing⁴⁰. The unigram “canal” is used to test what the print record measures. For the semantic sequence we retrieved sixteen bigrams naming coal’s mode of use and use eight: “coal barge”, “coal wharf”, “coal boat” and “canal boat” for coal moved by water; “steam engine” and “steam power” for coal burned in engines; “coal field” as the geologist’s term; and “fire engine”, the eighteenth-century name for an atmospheric engine, as a check on terminological drift. “Steam boat” was retrieved and excluded

because it names water transport as much as steam. Each bigram is indexed to its own 1850 frequency and the groups are equal-weighted averages of their members, so that a single common term cannot dominate; a further five-year centred mean is applied before thresholds are read.

3.5 Empirical strategy

Our questions are about timing, sequence and incidence within one national economy over 170 years. The sample is small, the series are smooth and trending, and the treatment is itself a slowly accumulating stock. We match the tools to those facts, and we use linear methods throughout: with 130 annual observations, flexible machine-learning estimators fit the time trend closely enough to leave no treatment variation to estimate from.

Regime detection. For each British series we estimate a linear trend in logs and test for a single break in level and slope at an unknown date using the Andrews sup-F statistic with 15 per cent trimming, reporting the statistic against the Andrews critical value, and for two breaks by grid search with a minimum segment of twenty years⁴¹. We also fix the break at 1761 and test for a change in trend slope over 1700–1830 with Newey–West standard errors⁴²; since the data-chosen breaks fall later, this fixed date is a conservative pre-test. Agriculture and GDP per head are the placebo series: canals did not carry harvests to any great extent, and a precondition operating through population should leave income per head unchanged.

Dose-response. We regress the log of each output series on the canal stock, in thousands of miles, and a linear trend over 1700–1830, the period before railways. Because a cumulative stock is itself smooth and accelerating, we report three further specifications: a quadratic trend with a dummy for war years (1756–63, 1775–83, 1793–1815); a first-difference model with ten lags of new mileage, reporting the sum of the lag coefficients and the joint F-test; and a horse race that adds the print frequency of “steam” and the war dummy. We test the residuals of the level regressions for a unit root. We build mainly on coefficients that survive all specifications; where an outcome survives most but not all, we say so.

Predetermined doses. Canals were promoted where trade was growing, so the contemporaneous stock is not exogenous. We therefore also use two doses fixed before the outcome window: the completion-based stock lagged ten and fifteen years, and the cumulative number of canals *authorised* by Act of Parliament, from Priestley, lagged ten years. An Act precedes completion by roughly a decade and cannot respond to output after it is passed. These doses remove

contemporaneous feedback; they do not remove anticipation, and we test for it directly by regressing new authorisations on past output growth and by adding the outcome's own past growth to the dose regressions.

Sequencing by local projection. To test the chain water $\rightarrow$ coal $\rightarrow$ steam $\rightarrow$ income per head, we estimate local projections⁴³: the change in the log outcome over horizons of one to twenty years is regressed on the regressor of interest, the outcome's own level and a trend, with Newey–West errors whose bandwidth is the larger of ten years and the horizon. Agriculture is again the placebo. Steam equations start in 1760, when the horsepower series begins; the income equations are estimated on 1760–1830, 1760–1870 and 1830–1870, with an interaction of horsepower and a post-1830 indicator, since the thesis predicts that steam's per-capita effect appears only after it becomes the majority power source. With eight outcomes, four specifications and four reported horizons, we treat a Bonferroni threshold of about 0.002 as the family-wise bar and say which results clear it.

Semantic sequencing. For each bigram group we record the first year in which the five-year moving average reaches 10, 25 and 50 per cent of its 1850 level, and the year in which “steam engine” overtakes “fire engine” in print. Because the British sub-corpus is thin before 1780, we repeat the 25 per cent threshold with three-, five- and nine-year windows and with 1830 as well as 1850 as the reference year. We also re-estimate the steam links of the chain with the print frequency of “steam engine” in place of interpolated horsepower, as an independent, if noisier, measure of steam's diffusion.

Comparative cases and the cross-country test. The comparative cases are read as four checks on the rule the British result supplies: an economy with engineered water and no coal, one with coal and no engineered water, one that acquired both late, and one with neither in reach. We report growth between the Maddison benchmark years for each. To show why we do not go further, we also estimate the two-way fixed-effects difference-in-differences on Maddison GDP per head, in levels and logs, with 1761 as treatment date, for three control groups and three sample windows, with its event-study version and clustered and collapsed standard errors, and compare peak-to-trough drawdowns over 1785–1815 by country.

All code, the tidy data files, the Priestley parsing rules and the horsepower interpolation are in the replication repository named in the data availability statement, together with a mediation decomposition that is not identified once a quadratic trend is absorbed and is reported there as a negative result. The code and the text were prepared with the assistance of an AI tool under the author’s direction, as declared in the acknowledgements.

4. Results

4.1 Two growth regimes

Figure 1 plots the British series on a logarithmic scale with the two canal-building waves shaded. Panel (a) shows total output, population and income per head; panel (b) shows coal, industry and agriculture. The visual impression is of a fan opening after 1760: total output and population steepen together while income per head continues at its previous slope until the 1820s. Agriculture never steepens at all.

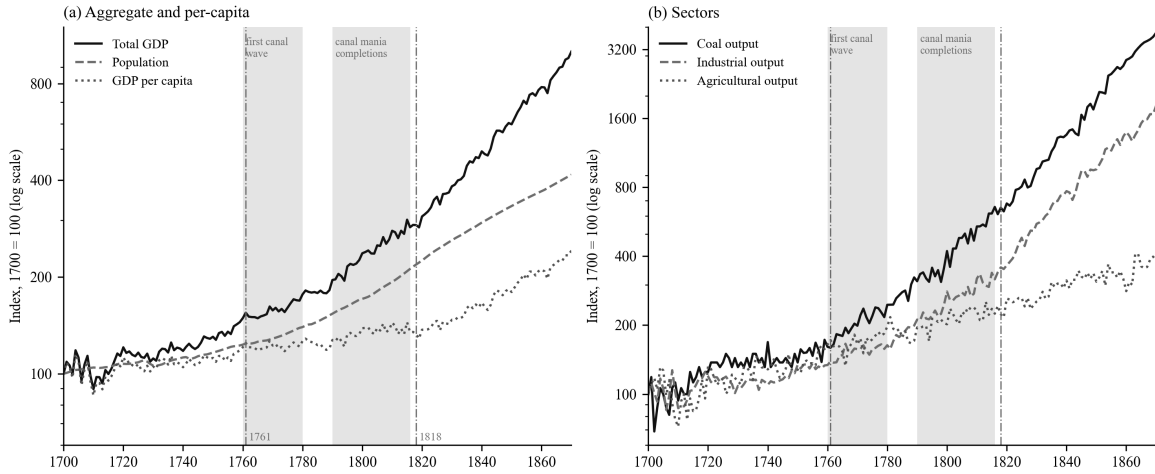


Figure 1: Britain’s two growth regimes. Annual indices, 1700 = 100, log scale. Shaded bands mark the first canal wave (1760–1780) and the canal-mania completions (1790–1816). Vertical lines at 1761 and 1818. Source: Bank of England millennium dataset⁴⁴.

Table 1 gives the trend growth rates by period. Between 1700–1760 and 1790–1815, the growth of total output nearly tripled, coal’s more than tripled and population’s quadrupled. Income per head grew at 0.25 per cent a year in the first period, 0.08 in 1760–1790 and 0.40 in 1790–1815. Agriculture is flat throughout.

Table 1: Trend growth of British series, per cent per year

Period	Total GDP	Industry	Coal	Services	Population	GDP per head	Agriculture
1700–1760	0.55	0.47	0.87	0.50	0.29	0.25	0.75
1760–1790	0.82	1.16	2.05	0.83	0.74	0.08	0.60
1790–1815	1.58	1.81	2.85	2.02	1.18	0.40	0.81
1815–1830	1.98	3.64	2.84	1.62	1.49	0.50	0.63
1830–1870	2.35	2.85	3.57	2.62	1.16	1.19	0.78

Source: own calculations from the Bank of England millennium dataset⁴⁵. Slopes of log-linear trends fitted within each window.

Table 2 formalises the comparison. Fixing the break at 1761 and estimating over 1700–1830, the trend slope rises by between 0.83 (population) and 3.24 (iron) percentage points a year for every canal-served series, all with p-values below 0.01 under Newey–West errors. The slope of income per head changes by 0.02 points and that of agriculture by -0.04 , neither distinguishable from zero. The fixed date is a pre-test, so the table also reports the break the data choose. The single best break falls between 1777 (population) and 1792 (total output) for every canal-served series, with sup-F statistics far above the Andrews critical value, and at 1818 for income per head. Agriculture’s best break is not significant: there is none. Allowing two breaks places the first in 1774–1775 for total output, industry and services and the second in 1818–1823 for output and industry; coal’s first break falls earlier, in 1741, and its second in 1798, which we return to below. The data pick out the two regimes without being told where to look.

Table 2: Change in trend slope at 1761 and data-chosen break dates

Series	Trend before 1761, % per year	Change after 1761, points per year	p	Best single break (sup-F)	Best two breaks
Total GDP	0.52	+0.85	<0.01	1792 (778)	1775, 1818
Industry	0.39	+1.42	<0.01	1789 (909)	1774, 1823
Coal	0.77	+1.62	<0.01	1784 (667)	1741, 1798
Iron	0.26	+3.24	<0.01	1786 (1,294)	1786, 1815
Textiles	0.69	+1.51	<0.01	1779 (240)	1728, 1817
Services	0.44	+1.07	<0.01	1786 (927)	1775, 1844
Population	0.23	+0.83	<0.01	1777 (3,052)	1730, 1783
<i>GDP per head</i>	0.29	+0.02	0.82	1818 (270)	1720, 1818
<i>Agriculture</i>	0.82	-0.04	0.84	none (4.2, not significant)	1720, 1757

Slope-change regressions on 1700–1830 with Newey–West (10 lags) errors. Break searches on 1700–1870: Andrews sup-F with 15 per cent trimming (5 per cent critical value approximately 11.8 for two parameters) and a two-break grid search with a minimum segment of 20 years, which cannot place a break after 1850.

The arithmetic of the first regime is simple. Between 1760 and 1815 the growth of total output and the growth of population each rose by about a percentage point a year: the economy grew faster and fed more people at the same income, for half a century, while coal output per head doubled, from an index of 100 in 1700 to 202 in 1790 and 246 in 1800. Coal's early break in 1741 is a reminder that the coal trade did not begin with canals⁴⁶; what the canal era added was inland coal, and the second break in 1798 dates it.

4.2 The canal network as a dose

Figure 2 shows the canal series. Panel (a) gives miles opened per decade: 117 in the 1760s, 353 in the 1770s, a lull of 83 in the 1780s, then 548 in the 1790s, 479 in the 1800s and 344 in the 1810s. Panel (b) gives the cumulative stock, from 218 miles in 1760 to 772 in 1790, 1,487 in 1800 and 2,320 in 1830. Panel (c) gives coal output per head. The correspondence between the two waves and the two steepenings of coal per head is visible; the regressions ask whether it survives detrending.

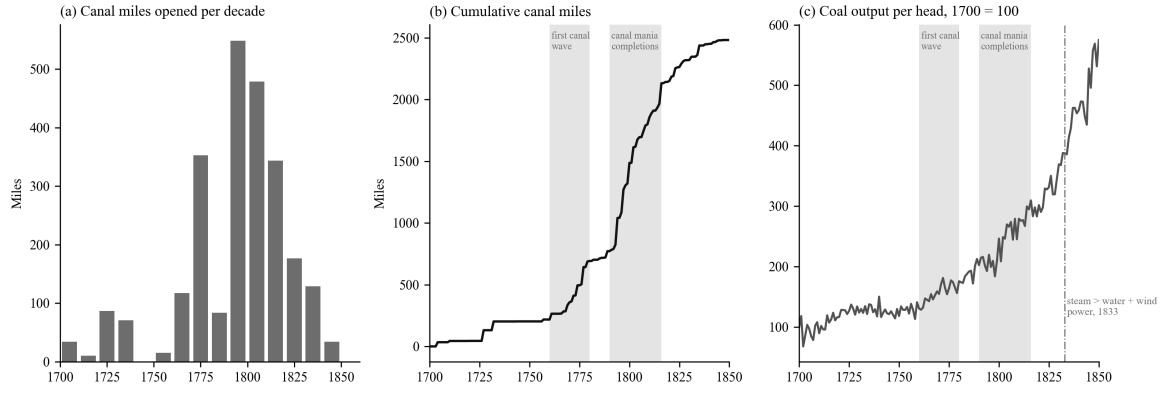


Figure 2: The canal network and coal. (a) Canal miles opened per decade, 155 canals by completion year. (b) Cumulative canal miles since 1700. (c) Coal output per head, 1700 = 100, with the year steam overtook water and wind as a source of stationary power. Sources: canal reference table described in the data section; Bank of England millennium dataset⁴⁷; horsepower benchmarks from Kanefsky as tabulated in Crafts⁴⁸.

Table 3 reports the dose-response regressions over 1700–1830. In the simplest specification, log output on a linear trend and the canal stock, a thousand miles of canal is associated with 47 per cent more coal, 40 per cent more industrial output, 99 per cent more iron, 32 per cent more services and 24 per cent more population, and with no change in income per head or agriculture. The sceptical specifications thin this out. With a quadratic trend and a war dummy, coal, iron and services retain large and significant coefficients; population is marginal; industry and total output are indistinguishable from the trend. In first differences with ten lags of new mileage, coal, industry and population respond, total output and income per head do not, and agriculture’s response is negative. In a horse race against the print frequency of “steam”, coal’s canal coefficient is 41 per cent against an insignificant 13 for steam. A horse race against installed steam horsepower is not informative: over 1760–1830 the interpolated horsepower series and the canal stock correlate at 0.98, so partial coefficients apportion a common trend. The ordering evidence must come from timing.

Table 3: Canal stock and British output, 1700–1830 (per 1,000 canal miles)

Outcome (log)	Linear trend	Quadratic trend + war	First differences, 10-year cumulative	Horse race: canal	Horse race: “steam” in print
Coal	+46.5% (p<0.001)	+29.8% (p=0.001)	+27.0% (p<0.001)	+40.8% (p<0.001)	+12.6% (p=0.08)
Industry	+40.1% (p<0.001)	−0.9% (p=0.89)	+14.3% (p<0.001)	+28.8% (p<0.001)	+30.4% (p=0.003)
Iron	+99.3% (p<0.001)	+61.2% (p=0.013)	—	—	—
Services	+32.3% (p<0.001)	+12.2% (p=0.002)	—	—	—
Population	+24.0% (p<0.001)	+4.1% (p=0.07)	+16.6% (p<0.001)	+18.6% (p<0.001)	+15.0% (p<0.001)
Total GDP	+24.7% (p<0.001)	+2.2% (p=0.52)	+4.2% (p=0.25)	+18.8% (p<0.001)	+19.0% (p<0.001)
<i>GDP per head</i>	+0.8% (p=0.69)	−1.9% (p=0.61)	−12.4% (p=0.33)	+0.2% (p=0.94)	+4.0% (p=0.08)
<i>Agriculture</i>	−0.5% (p=0.90)	−5.1% (p=0.56)	−18.8% (p<0.001)	−1.5% (p=0.72)	+7.3% (p=0.07)

Newey–West errors with 10 lags. The first-difference column reports the sum of the coefficients on lags 0–10 of new mileage, with the p-value of the joint F-test that all eleven are zero. The steam regressor is the Google Books frequency of “steam”, indexed to 1830. The war dummy covers 1756–63, 1775–83 and 1793–1815. An augmented Dickey–Fuller test rejects a unit root in the residuals of the linear-trend regressions at 5 per cent for every outcome except population ($p = 0.24$).

Two features of Table 3 matter. Coal is the robust channel, and it is the channel the mechanism predicts, since the canals were dug to move it. And the placebo rows behave: income per head does not respond in any specification, and agriculture’s only significant coefficient is negative, in first differences, the structural shift away from farming rather than an effect on it.

The reverse regression is not empty: past growth in coal, total output and population predicts subsequent canal openings, and past coal growth predicts new parliamentary authorisations with a joint p-value below 0.001. Canals were built where demand was growing. The predetermined doses address the contemporaneous part of this. With the completion stock lagged ten or fifteen years the pattern of Table 3 is unchanged: coal 48 to 49 per cent per thousand miles, industry 40 to 43, population 24 to 25, income per head nothing. With the authorisation count lagged ten

years, each ten canals authorised is followed by 12 per cent more coal, 10 per cent more industrial output and 6 per cent more population, again with no response in income per head; adding the outcome's own growth over the preceding decade leaves these at 11, 9 and 5 per cent, all significant at 1 per cent. Under a quadratic trend the authorisation dose falls to 4 per cent for coal, significant only at the 7 per cent level. In local projections it predicts coal at every horizon and does not predict population, income per head or agriculture. The dose is not exogenous; what we claim is that its timing and incidence, under contemporaneous and predetermined measures alike, are those of a precondition.

4.3 The precondition chain

Figure 3 reports the local projections that test the sequence directly. Each panel shows the cumulative response of the outcome, in log points, to a unit of the regressor, at horizons of one to twenty years, with 95 per cent Newey–West bands whose bandwidth grows with the horizon. The responses are predictive, not causal.

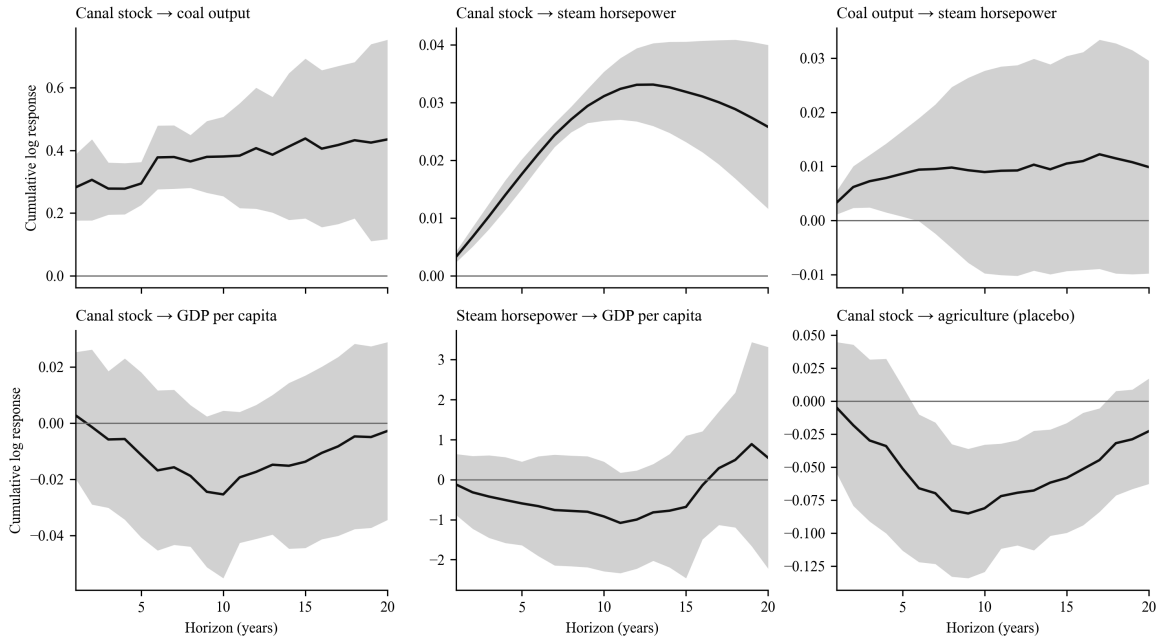


Figure 3: Local projections along the precondition chain. Cumulative log response at horizons 1–20 years, per 1,000 canal miles or per log point of the regressor, controlling for the outcome's level and a trend. Samples 1700–1830 where the canal stock is the regressor and 1760–1830 where steam horsepower enters, including the steam-to-income panel. Shaded: 95 per cent bands with Newey–West bandwidth equal to the larger of ten years and the horizon.

The top-left panel is the first link: a thousand miles of canal raises coal output by 29 per cent after five years, 38 per cent after ten and 44 per cent after fifteen, with the band excluding zero at every horizon shown. Controlling for contemporaneous steam horsepower, estimated from 1760, the canal effect on coal remains at 21 per cent over five to ten years and is gone at fifteen to twenty. The top-middle panel, canal stock to steam horsepower, is positive and precisely estimated at every horizon, but because the horsepower series is interpolated between four benchmarks, the panel establishes that steam capacity rose after the network did, not how fast. The top-right panel, coal to steam, is positive at every horizon but significant only at five years. On the interpolated series that is as much as can be asked.

The bottom row is the test that distinguishes a precondition from a cause. Canal stock has no effect on income per head at any horizon (bottom-left); the point estimates are negative and the bands include zero throughout. Steam horsepower has no effect on income per head when the sample stops at 1830 (bottom-middle). On the 1760–1870 sample it does, at every horizon and with p-values below 0.002; estimated on 1830–1870 alone the response is again positive at every horizon with p-values below 0.001, and an interaction of horsepower with a post-1830 indicator is positive and significant. The elasticities, between 0.2 and 0.9 depending on sample and horizon, should not be read as magnitudes; the sign and the timing are the result. Steam's per-capita dividend is a post-1830 phenomenon. Canals raise coal and population (the population response is 3 to 7 per cent per thousand miles over five to twenty years, significant beyond five years at the family-wise threshold) and leave income per head where it was. The bottom-right panel, agriculture, is negative at intermediate horizons, again the composition effect, and returns to zero.

Because the horsepower series is interpolated, we re-estimate the steam links with the print frequency of “steam engine”, which correlates at 0.90 with interpolated horsepower but varies year to year. The ordering survives, more weakly: the canal stock predicts “steam engine” at ten years ($p = 0.06$) and coal output predicts it at twenty ($p = 0.003$). The print series predicts income per head negatively at fifteen years, a warning about the series rather than about steam: it records discussion of engines, not the power they supplied. The last link of the chain therefore rests on the horsepower benchmarks, the 1818 break in income per head and the 1833 crossover in installed power.

The chain, read from Figure 3 and Table 4, is: canals raise coal within a decade; steam capacity rises after the network and after coal, on a scale that only the benchmarks can date; steam raises income per head, but only once it is the majority power source. Water first, coal second, steam third, income last.

4.4 How much power was steam?

The precondition thesis requires that the first regime run on water rather than steam. Table 4 uses the Kanefsky benchmarks to check. In 1760 steam supplied about 6 per cent of Britain's installed stationary power; in 1800, when the first regime was thirty years old, 21 per cent; by 1820 about a third; in 1830 parity with water, at 47 per cent. Steam overtook water and wind combined in 1833, fifteen years after the per-capita break. Coal output per unit of installed steam horsepower fell from 100 to 37 between 1760 and 1800: coal grew far faster than the engines that could burn it, and most of it went to hearths, forges, kilns and salt pans, moved by water.

Table 4: Installed stationary power in Britain, thousands of horsepower

Year	Steam	Water	Wind	Steam share	Coal output per steam hp (1760 = 100)
1760	5	70	10	6%	100
1800	35	120	15	21%	37
1830	160	160	20	47%	19
1870	2,060	230	10	90%	6

Source: Kanefsky's estimates as tabulated in Crafts⁴⁹; coal output from the Bank of England millennium dataset⁵⁰. Shares from log-linear interpolation between benchmarks, wind included.

4.5 The semantic sequence

If coal moved by water before it burned in engines, the language of the period should say so. Table 5 records, for each term or group, the first year in which its smoothed frequency in the British corpus reached 10, 25 and 50 per cent of its 1850 level. "Canal" reaches a quarter of its mid-century frequency in 1763, the year after the Bridgewater opening; "coal barge" in 1781; "coal wharf" in 1800; "canal boat", a later coinage, in 1823. "Steam engine" reaches the same threshold in 1808 and "steam power" in 1826. The coal-by-water group as a whole crosses 25 per

cent in 1800, the coal-by-steam group in 1819. “Steam engine” overtakes “fire engine”, the older name for the same machine, in 1800, which dates the terminological consolidation of steam to the decade after the canal mania. Figure 4 plots the three indices.

Table 5: Year in which print frequency first reaches a share of its 1850 level (British English corpus, five-year mean)

Term or group	10%	25%	50%
“canal”	1743	1763	1809
“coal barge”	1753	1781	1783
“coal wharf”	1774	1800	1815
“canal boat”	1812	1823	1825
Coal-by-water group	1780	1800	1817
“steam engine”	1800	1808	1821
“steam power”	1821	1826	1831
Coal-by-steam group	1804	1819	1825
“coal field”	1788	1820	1831

Source: Google Books Ngram corpus, British English 2019 edition, three-year API smoothing and a five-year centred mean. Groups are equal-weighted averages of member terms each indexed to 1850; the coal-by-water group also contains “coal boat”, which is too rare before 1800 to threshold on its own.

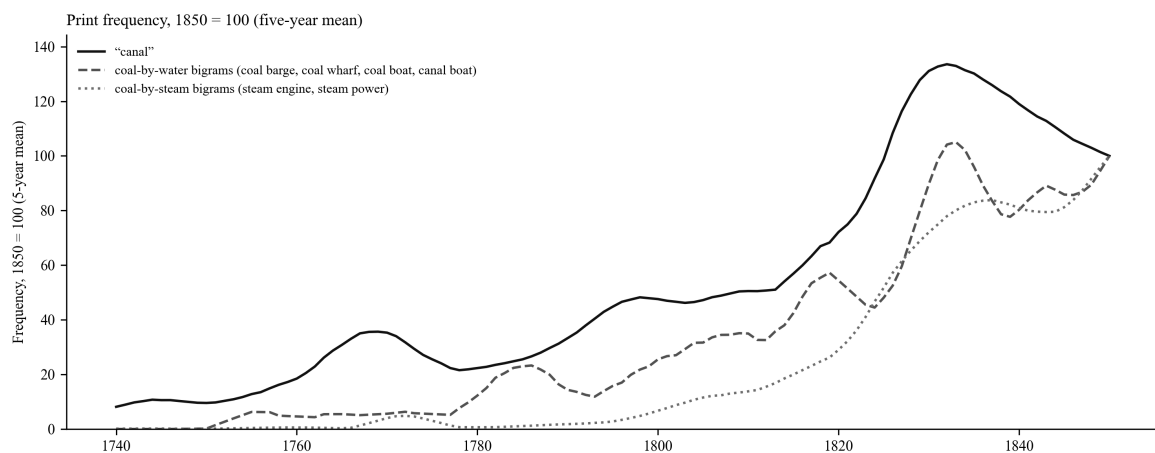


Figure 4: In print, coal travels by water before it is burned in engines. Five-year moving averages indexed to 1850 = 100. Coal-by-water: equal-weighted “coal barge”, “coal wharf”, “coal boat”, “canal boat”. Coal-by-steam: “steam engine”, “steam power”. Source: Google Books Ngram corpus, British English 2019 edition.

The ordering does not depend on the smoothing or the reference year. Across three-, five- and nine-year windows and with 1830 or 1850 as reference, “coal barge” crosses a quarter of its reference level in 1780–1781, “canal” in 1763–1766, “steam engine” in 1807–1809 and “steam power” in 1822–1826; the coal-by-water group crosses between 1785 and 1801 and the coal-by-steam group between 1814 and 1819 in every combination. “Coal wharf” is the one sensitive term, crossing in 1776 or 1800 depending on the reference, and in both cases before steam. Read as decades, the order is stable.

The corpus also tells us what kind of thing the infrastructure vocabulary measures. The frequency of “canal” correlates at 0.91 with the cumulative mileage of canals in existence over 1740–1850 and at -0.04 with the mileage opened in the surrounding decade; linearly detrended, the first correlation falls to 0.23, and in first differences to 0.16. Print records the level of the network, which is written about every year it exists, not the rate of building. We use it only in the first sense.

4.6 The comparative cases: water without coal, coal without water

Tvedt’s thesis is comparative, and the British result gives it a form the comparative cases can be checked against. If engineered water was a precondition for a coal economy and coal the fuel of the second regime, then an economy with waterways and no coal should show at most the first regime; an economy with coal and no engineered water should show neither until the water arrives; and an economy that acquires both late should run the sequence compressed. Annual series joining output, canal building and installed power are not available elsewhere, so the check is at the Maddison benchmark years, where population is measured rather than interpolated, read alongside what the national literatures record about fuel and water. Table 6 gives growth between 1700 and 1820, before steam supplied a third of British power, and between 1820 and 1870.

Table 6: Growth between benchmark years, per cent

	GDP per head 1700–1820	Population 1700–1820	Total GDP 1700–1820	GDP per head 1820–1870
Britain (UK, Maddison)	+37	+148	+240	+76
Great Britain (Broadberry)	+37	+125	+209	+76
Germany	+35	+66	+124	+44
Belgium	+8	+72	+85	+82
Spain	+22	+39	+69	+16
France	+3	+46	+50	+65
Sweden	–29	+104	+44	+52
Netherlands	–11	+23	+9	+47
China	–43	+176	+58	+7

Sources: Maddison Project Database 2023⁵¹; Great Britain row from the Bank of England millennium dataset⁵². The 1700 figure for China is contested⁵³.

On either measure British output roughly tripled between 1700 and 1820, and Britain was the only economy in the panel to more than double its population while raising income per head; Germany doubled its output and France grew by half.

The Netherlands is the case of water without coal. By the 1660s it had built more than six hundred kilometres of *trekvaarten*, towpath canals with scheduled barge services, on top of the densest river and drainage network in Europe⁵⁴. It ran on peat, which gave it in the seventeenth century an energy supply per head no other European economy matched, and it imported the coal it burned from Newcastle and Liège⁵⁵. What it lacked, besides coal, was gradient: a country at sea level has water to float on and little to fall, and Dutch industry was powered by wind. Its total output grew 9 per cent between 1700 and 1820 and its income per head fell, for reasons the literature locates in trade, finance, wages and war rather than fuel⁵⁶. The case shows that water transport was not sufficient; it does not, on its own, show that coal was what was missing. What it shows is that the densest engineered waterways in Europe, with no cheap domestic fuel to carry and no falling water to drive mills, did not compound.

Belgium is the case of coal that waited for water, and it is the nearest thing in the panel to an experiment on the sequence. The Hainaut and Liège coalfields were worked throughout the eighteenth century, but the Mons–Condé canal that joined the Borinage to the Scheldt opened in 1818 and the Charleroi–Brussels canal in 1832⁵⁷. Belgian income per head grew 8 per cent between 1700 and 1820, less than Spain's, and 82 per cent between 1820 and 1870, more than Britain's, on the way to the densest railway network on the continent. Belgium ran Britain's sequence in a third of the time, and it began after the canals reached the pits. The second regime could arrive so quickly because steam, by 1820, was a technology that could be imported; the first regime could not be, because it was built into the ground.

China is the case of coal and water that did not meet. The Grand Canal, some 1,800 kilometres from Hangzhou to Beijing, was the longest artificial waterway in the world and a state conduit for grain⁵⁸. China's coal lay in the north-west, in Shanxi and Shaanxi, a thousand kilometres from the Jiangnan core where the population, the textile industry and the demand for fuel were⁵⁹. The rivers that might have joined them were of the wrong kind: the Yellow River carried silt rather than barges and changed course, and the monsoon gave rivers and canals alike a wet season and a dry one⁶⁰. The benchmarks record a 176 per cent rise in population and a 43 per cent fall in income per head between 1700 and 1820, though the 1700 level is contested⁶¹, and 7 per cent per-capita growth in the half-century after. This is what a Malthusian regime looks like, and it is the regime Britain left in aggregate in the 1770s and per head in 1818: an economy that had exhausted its organic frontier and could not reach its mineral one.

India we treat in a sentence, on Parthasarathi's evidence. Its coal lay in Bengal, at Raniganj, while its export industry, cotton, lay in Gujarat and on the Coromandel coast, and the rivers between ran with the monsoon⁶². The endowment was China's problem in a different geography.

None of these cases is a test in the sense that the British series are. They are four readings of one rule, and the rule is what the British result supplies: the economies with engineered water and no coal, or coal and no engineered water, did not leave the organic regime in this period; the two that had both, Britain from the 1770s and Belgium from the 1820s, did, in the order the precondition predicts. A comparative paper with regional series for the Low Countries and the Yangzi delta could turn the readings into tests.

4.7 Why we stop at benchmark years

The natural cross-country test of an eighteenth-century British take-off is a difference-in-differences on Maddison GDP per head, treating Britain from 1761 against continental controls. In logs, with France and the Netherlands as controls and year and country effects, it returns treatment coefficients of 0.08 to 0.31 with p-values below 0.001 for every control group and window we tried. Taken at face value they say that Britain pulled away from the continent from 1761. They should not be. Britain's gap to the controls is significantly negative in the 1700s and 1710s, so there is no parallel pre-trend to break. A sup-F search on the log gap to the Netherlands and France places the single break in 1807, and Figure 5 shows what happened then: Britain's income per head, indexed to 1790, stood at 109 in 1805 and 114 in 1815; the Netherlands' at 100 in 1805, 63 in 1808 and 72 in 1815. Underneath both, Britain's own income per head grew at 0.08 per cent a year between 1760 and 1790. There was no canal-era per-capita acceleration for any control group to reveal.

Table 7 gives the drawdowns. Between the late 1780s and the Napoleonic trough the Netherlands lost 44 per cent of its income per head, Portugal 48, Sweden 27, France 22 and Spain 13. Britain lost 1.4 per cent. The 1761 “treatment effect” measured against a continental control group is, to a first approximation, the difference between being blockaded and doing the blockading. That is why the comparative evidence above is read at benchmark years and for aggregate output: over 1790–1815 the per-capita series record the war, not the rivers.

Table 7: GDP per head, peak 1785–95 to trough 1795–1815 (Maddison Project Database 2023, 2011 international dollars)

Country	Peak	Trough	Drawdown	1815 relative to 1790
Britain (UK)	3,207 (1795)	3,161 (1798)	–1.4%	+13.6%
Netherlands	4,666 (1794)	2,632 (1808)	–43.6%	–28.3%
Portugal	2,063 (1785)	1,072 (1811)	–48.0%	–24.1%
Sweden	1,661 (1791)	1,221 (1809)	–26.5%	–11.9%
France	2,016 (1788)	1,580 (1801)	–21.7%	+1.3%
Spain	1,454 (1790)	1,265 (1811)	–13.0%	+3.2%
Germany	1,820 (1792)	1,725 (1805)	–5.2%	+8.1%

Source: Maddison Project Database 2023⁶³. The series labelled Britain is the United Kingdom, including Ireland.

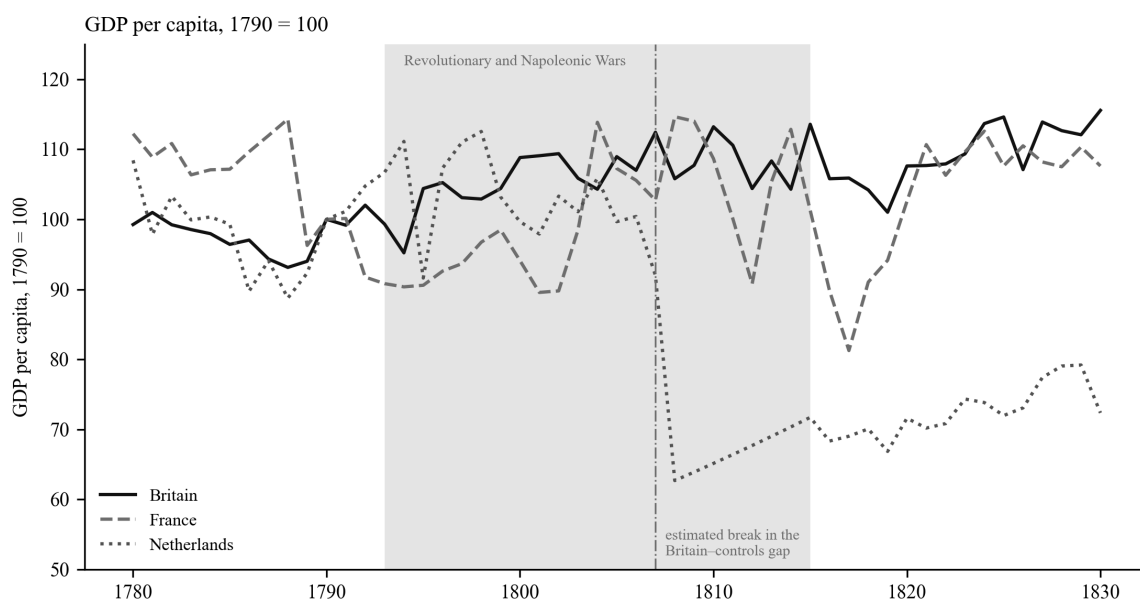


Figure 5: The 1807 “break” in the cross-country design is the Dutch collapse. GDP per head, 1790 = 100, for Britain, France and the Netherlands, with the Revolutionary and Napoleonic war years shaded and the estimated break in the Britain-controls gap marked. Source: Maddison Project Database 2023⁶⁴.

5. Discussion

5.1 Floated before fired

The results describe an economy that changed gear twice. The first change, in the 1770s and 1780s, was in the quantity of things: coal dug, iron smelted, goods made, people fed. The second, after 1818, was in the quantity per person. The first coincided with the building of the canal network and ran on water power and horse-drawn barges; the second coincided with steam becoming the majority power source. Coal is the hinge between them. Its output per head doubled during the first regime, when steam supplied between a twentieth and a fifth of installed power, and it is the one outcome whose response to the canal stock survives every specification, including the most demanding, where it retains significance only at the 7 per cent level.

This is Tvedt's sequence in the annual record. Britain's water system did not compete with coal; it delivered coal. The Bridgewater Canal existed to bring the Worsley pits to Manchester; the Grand Cross, begun in 1766 and completed when the Oxford Canal reached the Thames in 1790, joined the Trent, Mersey, Severn and Thames; the mania canals of the 1790s ran, disproportionately, from coalfields to towns. The sea-coal trade to London carried more tonnage throughout the century; what canals changed was where coal could go inland, and at what price⁶⁵. By 1800 coal was cheap far from the pithead, and the population had grown by half over the same forty years. We do not claim that canals caused that growth, which the demographic literature attributes to earlier marriage⁶⁶; we claim that the economy fed it at constant income, and that cheap inland coal and freight were part of how. The steam engine then found, ready-made, a market, a fuel supply, a factory system organised around water wheels, and in the canal company a legal template for raising capital by Act of Parliament⁶⁷. It had to improve on a prime mover that was, in 1800, three times its size.

5.2 What the first regime was, and was not

The first regime was not a rise in living standards. Income per head did not move, and the local projections show that canals did not move it. It was a rise in carrying capacity: the same income for half again as many people, sustained for two generations without the Malthusian check that had ended every earlier expansion. In Wrigley's terms this is not an escape from the organic economy, which by his definition Britain had begun to leave when it turned to coal for heat in the sixteenth century⁶⁸; it is the advanced organic economy working at full stretch, with mineral heat, organic power and engineered water. The escape from the ceiling in aggregate came before the escape per head, and it came before steam.

This also disposes of a false choice. Malm argues that the transition from water to steam power in the 1830s and 1840s was driven by the spatial and disciplinary needs of capital rather than by the exhaustion of water⁶⁹. Our results are consistent with that and add a prior stage to it: before steam could be chosen over water for its portability, water had to have created the markets and the fuel supply that made a portable prime mover worth having. Allen's induced-innovation story likewise needs coal to be cheap where engines were built⁷⁰; the canals are a large part of why it was.

5.3 The two clocks reconciled

The cliometric revision of the 1980s established that income per head grew slowly until 1830 and has been read as denying an eighteenth-century take-off⁷¹. Our results accept the revision and deny the reading. There was a take-off in the eighteenth century; it was an aggregate one, and it was absorbed by population. The canal historians and the growth accountants have been measuring different variables and talking past each other. Crafts's finding that steam's contribution to productivity growth is negligible before 1830 and large after 1850 is the second regime seen from the supply side⁷². Our finding that steam horsepower raises income per head only after 1830 is the same regime seen from the demand side.

One rival explanation of the 1818 break deserves to be named, because we have used it against the cross-country design: peace. Demobilisation, deflation and reopened trade could produce a per-capita break in 1816–1818 without steam. Absorbing a war dummy moves the estimated break from 1818 to 1816 and does not remove it, and the per-capita trend over 1816–1830, at 0.7 per cent a year, is well below the 1.2 per cent of 1830–1870. The break is real; its timing is over-determined; and the acceleration that follows it belongs to the steam decades.

5.4 Tvedt's question, answered as far as Britain can answer it

The comparative readings in section 4.6 let us say what the British test does for the global question. Tvedt's thesis, as we have restated it, is a rule about matching. Engineered water was necessary for a coal economy because, before railways, it was the only way to make coal cheap far from the pit; coal was necessary for the escape from the organic ceiling because water and wind could not be scaled. The British series establish the sequence in the one case we can watch year by year. In the benchmark cases no economy skipped a step. The Netherlands had the water and stayed organic. China had the coal and stayed organic. Belgium got the water in 1818 and left within a generation. Pomeranz's question, why Britain and not the Yangzi delta, and Tvedt's, why Britain and not the Netherlands, have on this reading one answer with the terms reversed: the delta lacked water that could reach its coal, and the Netherlands lacked coal its water could reach⁷³.

What the rule does not do is explain why Britain matched the two when it did. That Britain's rivers could be engineered at low cost is Tvedt's point; that they were engineered in the 1760s rather than the 1660s is a question about landowners, Parliament and the price of coal in

Manchester, on which national series are silent⁷⁴. A rule about necessary conditions leaves the timing of the sufficient ones open, and four cases cannot exclude other conditions, institutions or Atlantic trade among them, that would also fit them.

5.5 A caution for historical difference-in-differences

The cross-country result is methodological. In the long eighteenth century the control group went to war on its own territory and the treatment group did not, so a comparison spanning 1790–1815 attributes the difference to whatever the British treatment happens to be. The remedies are the ones applied above: locate the break before interpreting the coefficient, examine the controls’ own series, and prefer within-country evidence where the treatment is national. The estimator is not at fault; the history is⁷⁵.

5.6 What the text can and cannot measure

The most useful text result is the negative one: the corpus records the level of the built environment, not its growth, so vocabulary is a fair proxy for stock and a poor one for investment. The bigram sequencing is on firmer ground because it asks about order rather than magnitude. “Coal barge” around 1781 and “coal wharf” around 1800 record coal on the water; “steam engine” around 1808 and “steam power” around 1826 record coal in the engine. The order in the language is the order in the economy, give or take a decade, and the technical-publishing bias of the corpus works in favour of exactly this kind of vocabulary⁷⁶.

6. Limitations

The argument rests on timing, sequence and incidence in a single national time series, and it carries the limitations of that design.

6.1 The dose is endogenous

Canals were built where coal and people were. The reverse regressions confirm it: past growth in coal, total output or population predicts subsequent canal openings, and past coal growth predicts new authorisations. The dose-response coefficients are therefore not causal effects of an

exogenous treatment. The predetermined doses remove contemporaneous feedback, and adding the outcome's own past growth leaves the authorisation coefficients intact, but nothing here removes anticipation: a promoter who expected coal demand to grow would seek an Act in advance of it. What the design establishes is that the canal stock precedes coal output by five to twenty years under every measure of the dose and that income per head does not respond. Turnbull's regional evidence and Alvarez-Palau and co-authors' market-access estimates supply the cross-sectional identification this paper lacks⁷⁷; a county panel with terrain as the source of exogenous variation in canal access is the natural next step.

6.2 Trending series and multiple tests

Every series in the paper trends. The level regressions in Table 3 are exposed to spurious regression; the augmented Dickey–Fuller test on their residuals rejects a unit root at 5 per cent for every outcome except population, so the level evidence for population should be discounted and the first-difference and local-projection results relied on. The fixed 1761 break is a pre-test; the sup-F statistics are reported against the Andrews critical value, and agriculture's is not significant. With eight outcomes, four specifications and four horizons, a Bonferroni threshold of about 0.002 is the honest bar: coal, industry, population beyond five years and the post-1830 steam result clear it; the five-year population response, the authorisation dose under a quadratic trend and the coal-to-steam link do not.

6.3 The canal series is provisional

The mileage series is built from a published reference table of 155 canals by completion year. It omits river navigations improved before 1700 and minor branches, and it dates staged openings to their completion. Priestley's authorisation dates corroborate the two waves independently, but they were recovered by optical character recognition from an 1831 text and only 152 of 325 entries yielded a parseable year. It excludes the Scottish canals, some 150 miles by 1830, against output series for Great Britain. A definitive series would come from the Cambridge Group's sectional opening dates, which are not publicly deposited⁷⁸. Our magnitudes per thousand miles should be read with that in mind; the timing is less fragile.

6.4 Steam and water horsepower are interpolated

Kanefsky's estimates exist at four dates within our sample. The interpolated series are log-linear between them, so regressions that use them are identified from three kinks: the local projections involving steam establish ordering across benchmarks rather than annual dynamics, their standard errors understate the uncertainty, and the elasticities of industry and income per head with respect to steam should not be read as magnitudes. The same interpolation makes water horsepower collinear with the canal stock over 1760–1830 ($r = 0.98$), so the two halves of Tvedt's water thesis, transport and power, cannot be separated with these data; our dose is transport only. The print frequency of "steam engine" confirms the canal-to-steam ordering weakly and cannot check the steam-to-income link; engine counts by decade would tighten both⁷⁹. That steam raises income per head only after 1830 does not depend on the interpolation: it follows from the 1818 break, the 1833 crossover in installed power, and estimation on 1830–1870 alone.

6.5 Agriculture is an imperfect placebo

Agriculture's response to the canal stock is negative and significant at ten to fifteen years in the local projections and in first differences. We interpret this as compositional: labour and capital moved out of farming as canal-served sectors grew. It is not, strictly, a null result, and a reader who prefers a cleaner placebo can substitute income per head, which is null in every specification.

6.6 Text frequencies measure print, not speech

The Google Books corpus over-represents technical and legal publishing, and its British-English sub-corpus is small in the eighteenth century, so bigram frequencies before 1770 are noisy; the jump of "coal barge" from a quarter to half of its reference level in two years is a count spike, not an event. We therefore use thresholds of a reference level rather than absolute frequencies, and groups of terms rather than single terms. The ordering of water terms before steam terms holds for every smoothing window and reference year we tried; individual crossing years move by one or two years, and "coal wharf" by more, so the years should be read as decades. "Fire engine" also denotes a fire pump, which is why we use it only to date the terminological consolidation of "steam engine".

6.7 The comparative cases are readings, not tests

The Dutch, Belgian, Chinese and Indian cases rest on Maddison benchmarks and on the national literatures, not on series comparable to the British ones. Annual Dutch output by sector exists from the fourteenth century, but the trekvaart network was complete by 1665 and supplies no varying canal dose. Population in the Maddison database is interpolated between 1700 and 1820, so cross-country regressions on total output over that period are not legitimate and we have not run them; Table 6 is descriptive, and its Maddison row for Britain is the United Kingdom including Ireland. The trekvaart mileage, the Belgian canal dates and the location of Chinese and Indian coal are taken from the works cited, and the 1700 Chinese benchmark is contested. The war-drawdown result is a property of the controls and so is robust to their choice, but it means that cross-country per-capita evidence for a British take-off before 1815 is, in our reading, unrecoverable from the Maddison series with any difference-in-differences design.

7. Conclusion

Why England, and not the Netherlands, China or India? The answer this paper can give is narrower than Tvedt's and better measured. Britain industrialised on two clocks. The first, set in the 1770s, measured how much the economy produced and how many people it fed; the second, set around 1818, measured how much each of them had. The first ran on water. Canals carried the coal inland, water wheels drove the mills, and the aggregate economy accelerated while steam supplied a twentieth and then a fifth of its power. The second ran on steam, and it was steam that finally raised income per head. Coal connects the two: its output per head doubled in the first regime, when most of it was burned in hearths, forges and kilns rather than engines, and it is the one variable whose response to the canal network survives every specification we tried.

This is what a precondition looks like in data. Water infrastructure did not cause the Industrial Revolution in the sense of raising British incomes; it did not, and the local projections say so. It made the coal economy possible by making coal cheap far from the pit, and it helped feed, at constant income for two generations, the population that would work in the steam economy. Read as a rule, the British sequence says that neither engineered water nor coal was sufficient and that the fossil economy began where the two met. The Netherlands, with the water and without the coal, stayed in the organic regime; China, with the coal and without water that could reach it,

stayed there too, and India with it; Belgium, which had the coal and got the water in 1818, ran Britain's sequence in a third of the time. Four benchmark cases cannot prove a rule, but they can fail to contradict it, and they do.

Three things follow for how the period is studied. The distinction between aggregate and per-capita growth should be made explicit in any account of the eighteenth century, because the two chronologies that have divided the literature are chronologies of different variables. Cross-country difference-in-differences spanning the Revolutionary and Napoleonic wars should be read with the controls' own histories in view, because the largest economic event of the period happened to the control group. And the digitised print record, used with care, can date the use of the built environment even where it cannot measure its growth; in this case it dates the coal barge a generation before the steam engine.

The fossil economy was floated before it was fired. That is a claim about order; the order is in the British data, and the comparative record is consistent with it.

Data Availability Statement

All code and data required to reproduce the analyses are publicly available at [\[REPOSITORY URL REDACTED FOR REVIEW\]](#). British sectoral output, population and capital stock are from the Bank of England's *A Millennium of Macroeconomic Data for the UK*⁸⁰, which reproduces Broadberry et al. (2015). Cross-country GDP per head and population are from the Maddison Project Database 2023⁸¹. Installed horsepower benchmarks are from Kanefsky (1979) as reported in Crafts (2004). Canal completion years and lengths are compiled from published reference tables and parliamentary authorisation years are parsed from the digitised text of Priestley (1831); both series are included in the repository with their construction scripts. Word and phrase frequencies are from the Google Books Ngram Corpus, British English 2019 edition. A self-contained replication package is available as a supplementary archive.

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